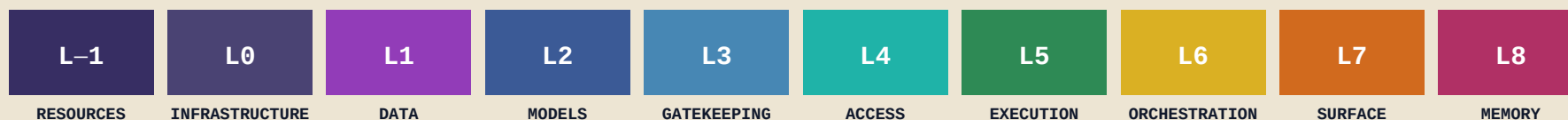


The supply chain of intelligence.

Ten layers. Fifty sublayers. Eighteen defensible positions.

The only framework that tells you where value accrues — and who can fire you.



By Anand Arivukkarasu · Ex-Meta (Instagram) Product Leader & AI Product Architect

Read it. Print it. Pin it to your roadmap wall.

01 One layer per page

Each page covers one layer of the stack with its color, definition, sublayers, and how product leaders use it.

02 Star = defensible

★ marks the 18 sublayers where structural moats can be built. Memorize these.

03 Read top to bottom

L-1 → L8. Below the line is foundation. Above the line is where intelligence compounds.

04 Map your product

On the final page, place your product, your competitors, and your moats on the stack.

05 Re-use the chips

Layer colors and labels are canonical. Use them in your own decks for instant recognition.

L-1

Resources.

The mines of the stack.

Energy, water, fabs, materials, skilled trades.

FIVE SUBLAYERS

Energy

Thermal & Water

Fabrication & Foundry

Critical Materials

Skilled Trades

HOW PRODUCT LEADERS USE THIS LAYER

*Map your bottleneck exposure:
which L-1 input (power, fabs,
materials) caps your growth?*

★ = STRUCTURAL MOAT

Sublayers marked ★ are positions where competitors cannot easily replicate you. Own at least one per product.

L0

Infrastructure.

The shovels of the stack.

Chips, data centers, networking, cloud, edge.

FIVE SUBLAYERS

Silicon (GPU/TPU)

Data Centers

Interconnect Fabric

Compute Access Cloud

Edge & On-Device

HOW PRODUCT LEADERS USE THIS LAYER

Decide build-vs-rent compute. If burn rate is dominated by L0, you're a tenant — not an owner.

★ = STRUCTURAL MOAT

Sublayers marked ★ are positions where competitors cannot easily replicate you. Own at least one per product.

L1

Data.

The ore of the stack.

What raw input do you have that nobody else can get?

FIVE SUBLAYERS

Public & Open Data

Proprietary Data ★

Behavioral Data ★

Outcome Data ★

Synthetic Data

HOW PRODUCT LEADERS USE THIS LAYER

Audit your proprietary data. If your moat is public data, you don't have a moat.

★ = STRUCTURAL MOAT

Sublayers marked ★ are positions where competitors cannot easily replicate you. Own at least one per product.

L2

Models.

The smelter of the stack.

Intelligence refinement. Rent early, build at scale.

FIVE SUBLAYERS

Foundation Models

Specialized Models

Embedding & Retrieval

Routing & Composition

Reasoning Models

HOW PRODUCT LEADERS USE THIS LAYER

*Pick model layer as a portfolio.
Never single-source. Track L2 →
L5 absorption risk.*

★ = STRUCTURAL MOAT

Sublayers marked ★ are positions where competitors cannot easily replicate you. Own at least one per product.

L3

Gatekeeping.

The hallmark of the stack.

Trust, acceptance, approval. Can the system be allowed in?

FIVE SUBLAYERS

Compliance Gates

Quality / Evals

Safety & Security

Editorial Gates ★

Distribution Gates ★

HOW PRODUCT LEADERS USE THIS LAYER

Treat compliance + evals as product, not paperwork. L3 is the slowest moat to build.

★ = STRUCTURAL MOAT

Sublayers marked ★ are positions where competitors cannot easily replicate you. Own at least one per product.

L4

Access.

The railroad of the stack.

Connectivity, permissions, integrations — the pipes.

FIVE SUBLAYERS

API & Integration

Agent Protocols ★

Access Governance

Real-Time Infra

Agent Identity ★

HOW PRODUCT LEADERS USE THIS LAYER

Invest in railroad tracks. Deep integrations = switching cost = survival.

★ = STRUCTURAL MOAT

Sublayers marked ★ are positions where competitors cannot easily replicate you. Own at least one per product.

L5

Execution.

The jeweler of the stack.

Applied skills and capabilities. Doing the actual work.

FIVE SUBLAYERS

Domain Execution ★

Decision Scaffolds ★

RAG Workflows

Operating Playbooks ★

Interactional Skills

HOW PRODUCT LEADERS USE THIS LAYER

Go deep, not wide. Generic execution dies; specialized execution compounds.

★ = **STRUCTURAL MOAT**

Sublayers marked ★ are positions where competitors cannot easily replicate you. Own at least one per product.

L6

Orchestration.

The store of the stack.

Workflow, routing, coordination — composing skills.

FIVE SUBLAYERS

Agent Loops

Human-in-the-Loop ★

Role Routing

Context & State

Runtime Assurance

HOW PRODUCT LEADERS USE THIS LAYER

Workflow ownership = customer ownership. Lose orchestration, you become a feature.

★ = STRUCTURAL MOAT

Sublayers marked ★ are positions where competitors cannot easily replicate you. Own at least one per product.

L7

Surface.

The wearing of the stack.

Interface and experience. How users meet the intelligence.

FIVE SUBLAYERS

Conversational

Visual / Media

Embedded & Copilot ★

Transaction Surface ★

Async & Ambient

HOW PRODUCT LEADERS USE THIS LAYER

Modality is commodity. Context (embedded / transactional) is the moat.

★ = STRUCTURAL MOAT

Sublayers marked ★ are positions where competitors cannot easily replicate you. Own at least one per product.

L8

Memory.

The ledger of the stack.

Retention, learning, compounding context.

FIVE SUBLAYERS

Session Memory

User Profiles

Network Learning ★

Institutional Knowledge ★

World Models ★

HOW PRODUCT LEADERS USE THIS LAYER

Build memory from day one. It's the only layer that gets stronger every day.

★ = STRUCTURAL MOAT

Sublayers marked ★ are positions where competitors cannot easily replicate you. Own at least one per product.

Pick three layers. Defend them ruthlessly.

No company wins all ten layers. The companies that survive pick three — usually one foundation layer, one execution layer, and one memory or distribution layer — and out-engineer everyone else there.

YOUR FOUNDATION

Layer: _____

Sublayer (★): _____

Why it's defensible:

YOUR EXECUTION

Layer: _____

Sublayer (★): _____

Why it's defensible:

YOUR COMPOUNDING LAYER

Layer: _____

Sublayer (★): _____

Why it's defensible:

Read the full framework + 50 sublayers at supplychainofai.com/framework